INCH-POUND
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MILITARY SPECIFICATION

CANNON, 105MM GUN: M68

CANNON, 105MM GUN: M68A1

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification supports the acquisition of two models of cannon. This specification includes the minimum essential Engineering and Packaging Requirements and the necessary Quality Assurance Provisions to determine that these requirements have been met.

1.2 Classification. The two models are:

Type I. Cannon 105MM Gun: M68
Type II. Cannon 105MM Gun: M68A1

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Armament Research, Development and Engineering Center, US Army Armament, Munitions and Chemical Command, ATTN: SMCAR-CCB-SC, Watervliet, NY 12189-4050 by using the self-addressed Standardization Document Improvement Proposal (DDForm 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1015

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specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-C-13931 - Cannon, General Specification for
MIL-P-14232 - Parts, Equipment and Tools for Army Materiel, Packaging of
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance, and Disposition of

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing and Marking
MIL-STD-2073-1 - DOD Materiel Procedures for Development & Application of Packaging Requirements

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.8)

US Army Armament Research, Development and Engineering Center

B7309991 - Magnetic Particle Inspection of Breech Rings (Wedge Type)
A7309992 - Magnetic Particle Inspection Criteria for Breech blocks (Wedge Type)
D8773565 - Gage, Tester, Insulation
B8775092 - Gage, Plug, Double End
A8775096 - Gage, Template
F8777451 - Gage, Continuity Tester Special
A8779071 - Instructions
B8769067 - Magnetic Particle Inspection of Cannon and Mount Components

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D8765947 - Cannon, 105MM: M68
P8765947 - Special Packaging Instruction, Cannon, 105MM: M68
C11578832 - Inspection Procedures and Criteria, Gun Tube Surface Indications
D11579700 - Cannon, 105MM: M68A1
P11579700 - Special Packaging Instruction, Cannon, 105MM: M68A1

PUBLICATIONS

Master LQAP 8765947 - List of Quality Assurance Provisions for Cannon, 105MM: M68
Master LQAP 11579700 - List of Quality Assurance Provisions for Cannon, 105MM: M68A1
QAP-APPENDIX-WVA - General Quality Assurance Provisions

(Copies of drawings, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the Procuring Contracting Officer.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

American Society for Testing and Materials (ASTM)

ASTM D 3951 - Commercial Packaging, Practice for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103-1137.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 General requirements. Cannon and component parts shall be in accordance with the applicable general drawing (D8765947 for M68 or D11579700 for M68A1), this specification, MIL-C-13931, and all referenced documents. Any conflicts are to be resolved by means of the order of precedence (see 2.3).

3.2 First article. The contractor shall submit a first article unless it is specifically waived in the contract (see 4.3 and 6.2). No first article requirements shall be waived without review and approval by the Procuring Contracting Officer (see 6.3).

3.3 Functional characteristics. Each cannon assembly shall perform the firing, pressure sustaining, and projectile launching function of its type gun, 105MM: M68 or M68A1. The breech mechanism, firing mechanism, and other moving parts shall function smoothly without interference, erratic movement or malfunction.

3.3.1 Breech mechanism. Camming of the operating crank during counter-recoil or rearward rotation of the operating lever rotates the breechblock crank rearward to lower the breechblock. When the breechblock nears the opened position, the lips of the extractors rotate sharply rearward to extract the cartridge case. The extractors then lock the breechblock in the opened position until tripped by the insertion of a cartridge. The breech mechanism with the closing spring adjustor set for minimum preload, shall then raise the breechblock to the closed (firing) position. Clockwise rotation of the adjustor shall increase the preload of the closing spring.

3.3.2 Firing pin retraction. With the initial opening movement of the breechblock, the firing pin retracts and remains behind the front face until the breechblock is returned to the closed position.

3.3.3 Firing pin protrusion. When the breech is closed, the protrusion of the firing pin shall be not less than 0.075 inch nor more than 0.115 inch.

3.3.4 Headspace. Headspace, the distance from the front face of the closed and locked breechblock to the breech face of the tube, shall be not less than 0.262 inch nor more than 0.282 inch.

3.3.5 Firing mechanism. The firing mechanism, consisting of the firing plunger, the firing contact, and the firing pin, shall meet the following requirements:

3.3.5.1 Continuity. The firing mechanism shall be capable of passing a current of at least 8 amps, and shall have a resistance

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of 1 ohm or less when using an applied test voltage of not more than 24 volts.

3.3.5.2 Dielectric strength (leakage). The dielectric strength (leakage) shall be such that no current in excess of 0.5 milliampere shall flow when the firing mechanism is subjected to an electrical potential of not less than 1000 volts, RMS, AC, 50/60 Hz for 30 seconds.

3.3.5.3 Insulation resistance. The firing mechanism shall have a minimum insulation resistance of 1 megohm when a potential of 500 volts DC is applied for 1 minute.

3.4 Performance.

3.4.1 High pressure resistance. The cannon shall be capable of withstanding a chamber pressure of 80,000 + 6000 pounds per square inch when tested as specified in 4.5.6.

3.4.2 Functioning. There shall be no impairment in the functioning of the components during and after the testing specified in 4.5.6.

3.4.3 Bore enlargement. The bore diameters, measured across the rifled lands after firing no more than six rounds, shall not exceed the before firing diameters by more than the amounts in TABLE I.

TABLE I. Allowable bore enlargement.

Distance forward of commencement of rifling	0.10 in.	1 in.	1 cal.	2 cals. to muzzle
Amount (in inches)	0.008	0.008	0.006	0.003

NOTE: Interpolate to determine the allowable enlargement between the distances listed.

There shall be no indication of abnormal wear, damage or deformation of the bore surface.

3.4.4 Material soundness. The cannon shall meet the requirements specified herein without evidence of cracks or fractures and shall comply with soundness requirements specified on the applicable drawings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and

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tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Production Qualification Test (PQT) (see 4.4).

c. Quality conformance inspection (see 4.5).

4.3 First article inspection. Unless otherwise specified in the contract, (see 6.2.1), first article cannon and specified parts thereof shall be inspected by the contractor and submitted to the Government for approval. Inspection shall be in accordance with applicable Quality Assurance Provisions (QAPs) and QAP-APPENDIX-WVA. QAPs are listed in Master LQAP 8765947 or in Master LQAP 11579700, as applicable. First article sample sizes shall be as specified in applicable item QAPs, or the contract, as applicable. First article cannon and parts thereof shall be representative of the production processes to be used during quantity production.

4.4 Production Qualification Test (PQT). When specified in the contract, (see 6.2.1), cannon shall be subjected to Production Qualification Testing by the Government to assure compliance with the requirements of this specification and any additional requirements listed in the contract. Production Qualification Testing procedures and sample size shall be as specified in the contract.

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4.5 Quality conformance inspection. Quality conformance inspection is listed in Table II.

TABLE II. Quality conformance inspection.

Category/Characteristic	Requirement Paragraph	Inspection Paragraph
Examinations		
Firing pin retraction	3.3.2	4.5.4.1
Firing pin protrusion	3.3.3	4.5.4.1
Headspace	3.3.4	4.5.4.2
Firing mechanism	3.3.5	4.5.4.3
Continuity	3.3.5.1	4.5.4.3.1
Dielectric strength (leakage)	3.3.5.2	4.5.4.3.2
Insulation resistance	3.3.5.3	4.5.4.3.3
Material soundness	3.4.4	4.5.4.4
Packaging	5.	4.5.4.5
Functional inspection	3.3	4.5.5
Breech mechanism	3.3.1	4.5.5.1
Extractors	3.3.1	4.5.5.2
Tests		
Proof acceptance (cannon)	3.4.1 to 3.4.4	4.5.6.1
Proof firing procedures (without sampling)	3.4.1 to 3.4.4	4.5.6.1.1
Proof firing procedures (with sampling)	3.4.1 to 3.4.4	4.5.6.1.2
Proof acceptance (breech mechanism)	3.4.1, 3.4.2, 3.4.4	4.5.6.2
General proof firing procedures	3.4.1 to 3.4.4	4.5.6.3

4.5.1 Inspection provisions. Cannon and parts thereof shall be inspected for conformance to requirements of this specification, MIL-C-13931, QAPs listed in Master LQAP 8765947, Master LQAP 11579700, as applicable, and QAP-APPENDIX-WVA.

4.5.2 Inspection approval stamp. The application of the inspection approval stamp shall be as specified in MIL-C-13931.

4.5.3 Inspection equipment.

4.5.3.1 Acquisition, maintenance and disposition. Unless otherwise specified (see 6.2.1), responsibility for acquisition, maintenance, and disposition of inspection equipment prescribed in applicable item QAPs, and for all other inspection equipment required to perform inspections prescribed by applicable specifications, shall be in accordance with MIL-I-45607.

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4.5.3.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it shall be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

4.5.4 Examinations.

4.5.4.1 Firing pin protrusion/retraction. Firing pin protrusion shall be inspected on each breech mechanism before assembly to the cannon tube. Inspection shall be performed by means of Gage A8775096 or an approved equivalent to determine compliance with 3.3.3. Firing pin retraction shall be visually observed to determine compliance with 3.3.2. Failure to comply with requirements shall be cause for rejection.

4.5.4.2 Headspace. Headspace shall be inspected on each cannon with Gage B8775092 or an approved equivalent to determine compliance with 3.3.4. Failure to comply with requirements shall be cause for rejection.

4.5.4.3 Firing mechanism. Firing mechanism electrical tests shall be performed on each cannon with the aid of gages F8777451, D8773565, or approved equivalents. For instructions and safety precautions pertaining to the use of these gages, refer to A8779071. Firing mechanism electrical tests shall consist of continuity, dielectric strength (leakage), and insulation resistance tests.

4.5.4.3.1 Continuity. Using gage F8777451 and appropriate test instruments, make connections to the point of the firing pin and to the external end of the firing plunger. An applied test voltage of not more than 24 volts shall be used. The current through the firing mechanism shall be measured and a value of less than 8 amps shall be cause for rejection. The resistance of the firing mechanisms shall be calculated (using the actual voltage drop across the firing circuit) or measured (after compensating for lead resistance), and a value of more than 1 ohm shall be cause for rejection.

4.5.4.3.2 Dielectric strength (leakage). After completion of the continuity test, gage D8773565 shall be connected to the external end of the firing plunger and to the breechblock. A 50/60 Hz potential of 1000 volts RMS shall be applied for 30 seconds. Leakage current exceeding 0.5 milliampere shall be cause for rejection.

4.5.4.3.3 Insulation resistance. After completion of the dielectric strength (leakage) test, gage D8773565 shall be connected to the external end of the firing plunger and to the

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breechblock. A potential of 500 volts DC shall be applied for one minute. Any mechanism having less than 1 megohm insulation resistance shall be cause for rejection.

4.5.4.4 Material soundness. All external and internal surfaces of the tube, breech ring, breechblock, and bore evacuator shall be examined by magnetic particle inspection in accordance with the procedures and acceptance criteria of the applicable General Data Drawings as listed below:

a. C11578832 for tubes.

b. B7309991 for breech rings.

c. A7309992 for breechblocks.

d. B8769067 for bore evacuators (with acceptance criteria of Class IV).

4.5.4.5 Packaging inspection. Packaging inspection for the level designated in the contract shall be performed in accordance with the applicable packaging data sheets and MIL-P-14232. Packaging failing to comply with the packaging requirements of Section 5 and the contract shall be cause for rejection.

4.5.5 Functional inspection. Functioning tests shall be conducted in accordance with MIL-C-13931 and as specified herein. A minimum of two complete cycles of operation shall be performed on each cannon.

4.5.5.1 Breech mechanism. Each breech mechanism shall be operated manually to determine smoothness of operation in compliance with 3.3.1. Action of the operating lever shall be observed to assure that the lever is locked when in the closed position. Any evidence of interference, erratic movement, or malfunction of any component shall be cause for rejection.

4.5.5.2 Extractors. With the cannon in a horizontal position, extraction and ejection shall be tested with an empty service cartridge case to determine compliance with 3.3.1.

4.5.6 Tests.

4.5.6.1 Proof acceptance of cannon or repair tubes. Each cannon or repair tube shall be proof fired on a non-sampling basis in accordance with 4.5.6.1.1 or on a sampling basis in accordance with 4.5.6.1.2. Proof firing shall be performed under the supervision of the Government at a proofing facility designated by the Government. Inspection before and after proof firing shall be in accordance with this specification and MIL-C-13931.

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4.5.6.1.1 Proof firing procedures (without sampling). Unless otherwise specified, proof firing of cannon or repair tubes shall be conducted without sampling and shall consist of the following:

a. The first cannon or repair tube manufactured under a contract shall be subjected to a proof firing sequence consisting of two rounds of ammunition with a specified maximum average pressure of 61,800 psi, followed by four rounds adjusted to produce a chamber pressure of 80,000 + 6,000 psi.

b. Each subsequent cannon or repair tube shall be proof fired with one round adjusted to produce a chamber pressure of 80,000 + 6,000 psi.

c. General proof firing procedures shall be followed in accordance with 4.5.6.3.

4.5.6.1.2 Proof firing procedures (with sampling). When specified (see 6.2.1), proof firing of cannon or repair tubes shall be performed on a sampling basis conducted as follows:

a. The Product Assurance Directorate at Watervliet Arsenal will designate the appropriate sampling plan parameters including the initial quantity (i) and sampling frequency (f), in the contract (see 6.2.1).

b. The first "i" quantity of cannon or repair tubes produced in a particular buy shall be proof fired. When the "i" quantity has been satisfied, sampling shall be put into effect using the "f" sampling frequency.

c. Each cannon or repair tube that is designated to be proof fired, shall be proof fired with one round adjusted to produce a chamber pressure of 80,000 + 6,000 psi. General proof firing procedures shall be followed in accordance with 4.5.6.3.

d. Each cannon not proof fired shall have the breech mechanisms proof tested (see 4.5.6.2) in the Product Assurance weapon test facility at Watervliet Arsenal or in a qualified weapon test facility approved by the Product Assurance Directorate of Watervliet Arsenal.

e. When specified in the contract (see 6.2.1), a weapon simulation test facility shall be qualified as follows: A sample of ten breech mechanisms of cannon to be sent to the proving ground for proof firing shall initially be simulation proof tested at the weapon simulation test facility. Qualification of the weapons test facility shall

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be determined by the Product Assurance Directorate at Watervliet Arsenal.

f. Proof fired cannon or repair tubes, and proof tested breech mechanisms shall be furnished with applicable data on DA Form 2408-4 (Weapon Record Data), as specified by the Product Assurance Directorate at Watervliet Arsenal.

4.5.6.2 Proof acceptance of breech mechanisms or repair breech mechanisms. The breech mechanism of each cannon not proof fired at a proving ground and each repair breech mechanism shall be proof tested in accordance with the following procedure:

a. Each breech mechanism and repair breech mechanism shall be tested by means of suitable hydrodynamic testing equipment by one application of a pressure within the proof pressure range of 80,000 + 6,000 psi.

b. Cycle time to achieve test pressure and release shall not exceed three minutes. Test pressure shall be immediately released upon its attainment.

(1) When difficulty is encountered in attaining test pressure, the number of attempts in which over 10 percent of the test pressure is attained shall not exceed five.

(2) If the cause of difficulty is not identified and corrected within five pressure cycles, the breech mechanism shall be subjected to proof firing as part of a cannon. A notification of this condition shall be submitted with the breech mechanism to the proving ground.

c. The peak test pressure shall be recorded for each test and attached to inspection records identifiable with the specific breech mechanism undergoing test.

d. After the pressure test, each breech mechanism shall be examined for looseness or interference of fit or other defects of workmanship and function (see 4.5.5).

e. Each breech mechanism shall be disassembled and examined for soundness of material. All components shall be visually examined to assure freedom from cracks and fractures. Each breech ring and breechblock shall be magnetic particle inspected (see 4.5.4.4).

4.5.6.3 General proof firing procedures. Proof firing shall be conducted under Government supervision. The general test procedure

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shall be as follows:

a. Cannon or repair tubes shall be proof fired in accordance with 4.5.6.1.1 or 4.5.6.1.2 as applicable.

b. A facility mount may be used if necessary.

c. Proof test cartridges shall consist of M900 slug projectiles assembled with standard M900 cartridge components using M43 propellant. An M11 crusher-type pressure gage will be loaded into the proof test cartridges to measure peak chamber pressure. The proof test cartridges shall be heated to 49 degrees C prior to firing. The chamber pressures shall be recorded.

d. After proof firing:

(1) The bore of each tube shall be examined for bore enlargement using an air gage, an electronic gage, or other approved gage. Enlargements exceeding the limits specified in TABLE I shall be cause for rejection (see 3.4.3).

(2) The bore surfaces of each tube shall be examined by means of a borescope (see 4.5.4.4).

(3) Each tube and bore evacuator shall be magnetic particle inspected (see 4.5.4.4).

(4) Each breech mechanism, including the extractors, shall be examined for looseness or interference of fit and other defects of workmanship and function (see 4.5.5).

(5) Each firing mechanism group shall be examined for looseness or interference of fit and other defects of workmanship and function (see 4.5.4.3).

(6) Each breech mechanism shall be disassembled and examined for soundness of material. All components shall be visually examined to assure freedom from cracks or fractures. Each breech ring and breechblock shall be magnetic particle inspected (see 4.5.4.4).

(7) Each cannon found to be defective at the proving ground shall be reported to the Product Assurance Directorate, Watervliet Arsenal, Watervliet, NY 12189-4050; and Benet Laboratories, Watervliet Arsenal, Watervliet, NY 12189-4050. The test agency shall forward a Test Incident Report (TIR) within 10 working days if incident is classified major, minor,

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or for information; critical must be reported by telephone within 12 hours.

5. PACKAGING

5.1 Cannon - Levels A, B and C. Cleaning, drying, preservation, unit packaging and packing of cannon shall be in accordance with the requirements of Special Packaging Instruction (SPI) AM8765947 and MIL-STD-2073-1 for Cannon, 105MM Gun M68 and SPI AM11579700 and MIL-STD-2073-1 for Canon, 105MM Gun M68A1 for the level of protection specified in the contract (see 6.2).

5.2 Repair parts - Level A, B and C. Cleaning, drying, preservation, unit packaging and packing of repair parts, shall be in accordance with the applicable repair part SPI and MIL-STD-2073-1 [or the applicable repair part PDS and MIL-P-14232] for the level of protection specified in the contract (see 6.2). When Level C requirements are not shown on the SPI/PDS, Level C packaging shall be in accordance with MIL-STD-1190.

5.2.1 Level X - industrial packaging. Cleaning, drying, preservation, unit packaging and packing of repair parts shall be in accordance with ASTM D 3951 when industrial packaging is specified in the contract (see 6.2).

5.2.1.1 Quantity in unit package (QUP). When Level C packaging in accordance with MIL-STD-1190 or industrial packaging of repair parts is specified in the contract, the QUP shall be the same as specified in the contract (see 6.2).

5.3 Marking of packages.

5.3.1 Levels A, B and C. Marking (including bar code marking) of the unit package, intermediate packages, and exterior shipping containers shall be in accordance with MIL-STD-129 (see 6.2).

5.3.2 Level X - industrial. Marking (including bar code marking) of the unit package, intermediate packages, and exterior shipping containers shall be in accordance with MIL-STD-129 (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The 105MM M68 is intended for use in most M60 series and the M48A5 Tank systems. The 105MM M68A1 is intended for use in most M60 series, M48A5, and M1 Tank systems.

6.2 Acquisition requirements. Acquisition documents must specify

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the following:

a. Title, number and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. Requirements for and the number of cannon, components thereof, or both to be submitted for first article by the contractor (see 3.2).

d. The examinations and tests to be performed by the contractor and the examinations and tests to be performed by the Government.

e. All inspection and test data that is required.

f. At whose expense a retest may be performed.

g. Any additional or extended examinations and tests beyond the scope of this specification.

h. Serial numbers for the cannon and repair tube assemblies.

i. Inspection equipment, responsibility for acquisition, maintenance, and disposition thereof, if other than as specified (see 4.5.3.1).

j. Availability of inspection equipment from the Government.

k. Extent of contractor's responsibility for Government furnished and for contractor-required final inspection equipment.

l. Applicable acceptance test procedures.

m. When a proof firing sampling plan is authorized, the proof firing sampling plan parameters: initial consecutive quantity (i), and sampling frequency (f) (see 4.5.6.1.2).

n. Under what conditions simulated firing may be performed, the test equipment available, and the procedures to be followed.

o. The levels of preservation, packaging, and packing required for cannon assemblies (see 5.1) and repair parts (see 5.2).

p. Marking and bar code marking requirements (see 5.3).

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q. When warranted, the application of MIL-Q-9858 or MIL-I-45208, as appropriate.

r. Unless otherwise specified, the application of MIL-I-45607 and MIL-STD-45662.

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 227.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title
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The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the Procuring Contracting Officer should provide specific guidance to offerers whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first lot of production items, a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.3. The Procuring Contracting Officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 International interest. Certain provisions of this specification are the subject of international standardization

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agreements, QSTAG-86. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

6.6 Waivers and deviations. The Procuring Contracting Officer shall coordinate all requests for waiver or deviation to this specification with Benet Laboratories and the Product Assurance Directorate at Watervliet Arsenal.

6.7 Subject term (key word) listing.

Cannon, 105MM, M68/M68A1
Gun, tank
M1 Tank, cannon for
M48A5 Tank, cannon for
M60 Tank, cannon for

6.8 Drawings. Drawings listed in Section 2 of this specification under the heading "U.S. Army Armament Research, Development and Engineering Center" (ARDEC) may also include drawings prepared by, and identified as Watervliet Arsenal, Ordnance Corps, Weapons Command, or U.S. Army Armament Research and Development Command, U.S. Army Armament, Munitions and Chemical Command, etc. Technical data originally prepared by these activities are now under the cognizance of ARDEC.

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1015-A332)

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	MIL-C-45504E (AR)	8 June 1992

3. DOCUMENT TITLE
CANNON, 105MM GUN: M68 CANNON, 105MM GUN: M68A1

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

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